Predictive Modeling of *Salmonella enteritidis* Growth Under Sinusoidal Temperature

Modified Gompertz Model & OLS Parameter Estimation

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目录

1 Introduction & Background

2 Model Description

3 Data Overview

4 Forward Problem

5 Parameter Selection

6 OLS Results

7 Model Diagnostics

8 Optimal Experimental Design

9 Bootstrapping

10 Conclusions

11 References

Introduction & Background

- *Salmonella enteritidis* is a major foodborne pathogen in egg products
- Temperature fluctuations during storage/transport affect microbial growth
- Reference: Gumudavelli et al. (2007), *J. Food Sci.* 72:M254–62 [1]

Objective

Use a modified Gompertz model to predict *Salmonella* growth under sinusoidal heating, estimate model parameters via OLS, and evaluate model adequacy.

Model Description

Primary Model (differential form)

$$\frac{d\log N}{dt} = \mu \cdot K \cdot C \cdot \exp(-K), \quad K = \exp\big(-\mu(t-M)\big) \tag{1}$$

Secondary Model (growth rate vs. temperature)

$$\mu = a(T - T_{\min})^2\big(1 - \exp[b(T - T_{\max})]\big) \tag{2}$$

- **Dependent variable**: $\log_{10} N(t)$ —log microbial concentration (log cfu/mL)
- **Independent variable**: t —time (hr)

Model Description —7 Candidate Parameters

Parameter	Symbol	Description	Units	Initial Guess
A	A	Initial log microbial number	$\log \mathrm{cfu/mL}$	$\log_{10}(400) \approx 2.60$
C	C	Upper–lower asymptote difference	$\log \mathrm{cfu/mL}$	11
M	M	Inflection point time	hr	7.5
a	a	Secondary model coefficient	$^\circ\mathrm{C}^{-2}\,\mathrm{hr}^{-1}$	0.000338
b	b	Secondary model coefficient	$^\circ\mathrm{C}^{-1}$	0.275
$T_{\min}$	$T_{\min}$	Minimum growth temperature	$^\circ\mathrm{C}$	6
$T_{\max}$	$T_{\max}$	Maximum growth temperature	$^\circ\mathrm{C}$	46.3

Data Overview

- Growth data: $\log_{10}(\text{CFU/mL})$ at multiple time points, $n = 20$
- Temperature: sinusoidal $T(t)$ recorded every minute
- Two independent time vectors

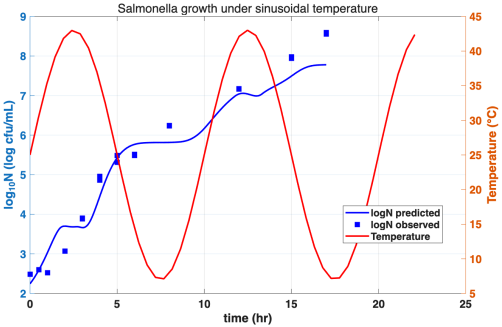

Fig 1. Dual-axis: $\log_{10} N$ and T vs. time

Forward Problem —Initial Guess

- All 7 parameter guesses from literature (Gumudavelli et al., 2007)
- ODE solved with ode45
- Temperature interpolated via `interp1`

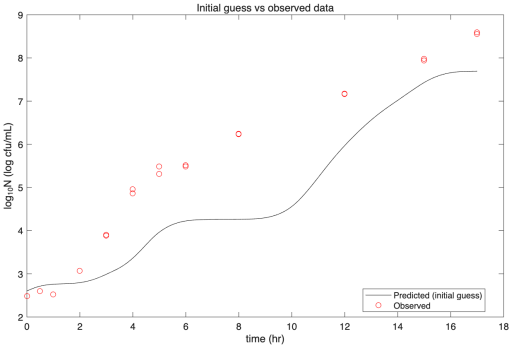

Fig 2. Initial guess vs. observed data

SSC Analysis —All 7 Parameters (Step 1)

Max|SSC|:

Param	Max|SSC|
A	2.61
C	5.11
M	5.45
a	2.20
b	**0.88**
$T_{\min}$	**0.83**
$T_{\max}$	9.02

Decision

$T_{\min}$ (0.83) and b (0.88): lowest SSC $\Rightarrow$ **fix**: $T_{\min} = 6\,^{\circ}\mathrm{C}$, $b = 0.275$.

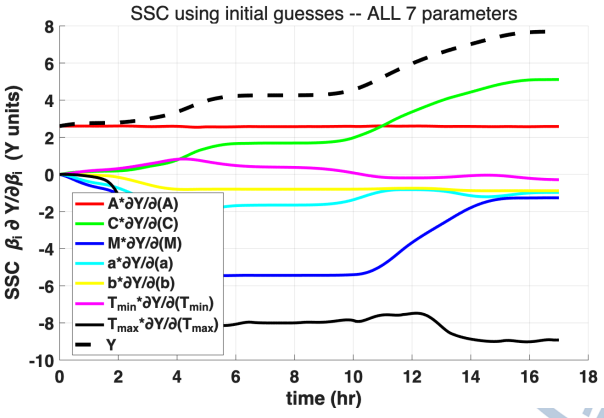

Fig 3. SSC for all 7 parameters

Round 1 —Estimate 5 Parameters $[A, C, M, a, T_{\max}]$

- Fixed: $T_{\min} = 6\,^{\circ}\mathrm{C},\ b = 0.275$
- Estimated: $A, C, M, a, T_{\max}$ $(p = 5)$

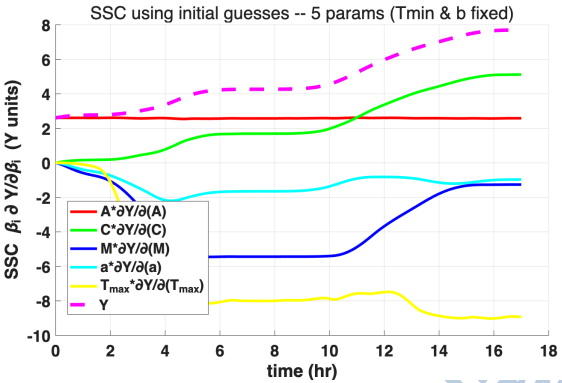

Fig 4. SSC with initial guesses (5 params)

Round 1 Results & Parameter Selection Step 2

- `nlinfit` OLS estimation (5 params): $[A, C, M, a, T_{\max}]$
- cond(**J**) $= 1.92 \times 10^6$ —still ill-conditioned
- Correlation matrix: M–a ($R = -0.994$), C–$T_{\max}$ ($R = -0.991$)

Key Finding

a and M are highly collinear: SSC shapes nearly proportional $\Rightarrow$ one must be fixed.

Decision

Fix a at Round 1 estimate $\Rightarrow$ reduce to **4 parameters** $[A, C, M, T_{\max}]$.

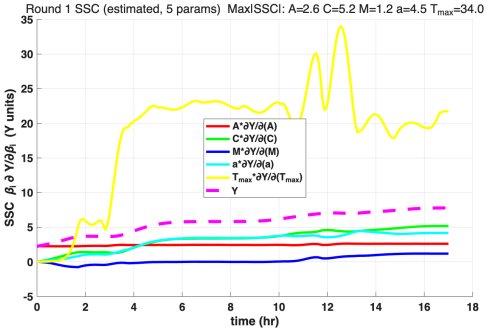

Fig 5. SSC at estimated values (5 params)

Parameter Selection Summary

Parameter	Max|SSC|	Identifiable?	Action	Value Used
A	2.61	✓Yes	**Estimate**	OLS result
C	5.11	✓Yes	**Estimate**	OLS result
M	5.45	✓Yes	**Estimate**	OLS result
$T_{\max}$	9.02	✓Yes	**Estimate**	OLS result
a	2.20	△ Collinear w/ M	Fix (Round 1 est.)	from Round 1
b	0.88	× Low SSC	Fix (literature)	$0.275\,^\circ\mathrm{C}^{-1}$
$T_{\min}$	0.83	× Lowest SSC	Fix (literature)	$6\,^\circ\mathrm{C}$

- **Stepwise elimination**: $7 \to 5$ (fix $T_{\min}$, b) $\to 4$ (fix a)
- **Final estimable parameters**: $[A, C, M, T_{\max}]$, $\mathrm{cond}(\mathbf{J}) = 293 \ll 10^6$
- SSC magnitude + collinearity check (ratio plot + $\mathrm{cond}(\mathbf{J})$) together determine parameter selection

OLS Inverse Problem —Final Results ($[A, C, M, T_{\max}]$)

Param	Estimate	σ	Rel Err	95% CI
A	2.244	0.122	5.5%	[1.985, 2.504]
C	13.387	0.519	3.9%	[12.29, 14.49]
M	3.529	0.023	0.7%	[3.479, 3.578]
$T_{\max}$	41.651	0.022	0.05%	[41.60, 41.70]

- Fixed: $T_{\min} = 6\,^\circ\mathrm{C}$, $b = 0.275$, $a = 3.6 \times 10^{-4}$ (Round 1 est.)
- cond$(\mathbf{J}) = 293 \ll 10^6$ RMSE $= 0.367$ relRMSE $= 6.3\%$ $R^2 = 0.972$
- Correlation: C–$T_{\max}$ ($R = -0.990$), A–M ($R = -0.956$) —correlated but distinguishable

Fitted Curve with Asymptotic CB and PB

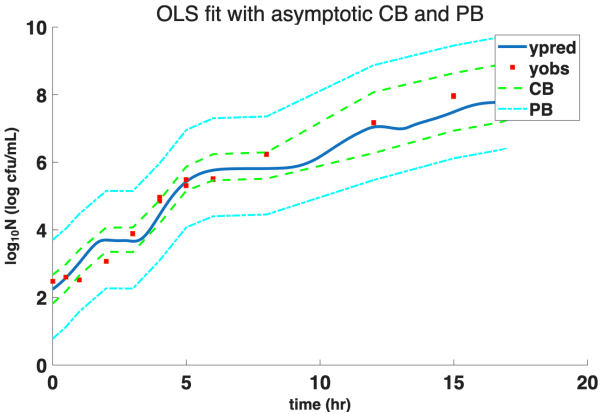

Fig 6. OLS fit with 95% simultaneous confidence bands (CB) and prediction bands (PB)

Residual Analysis

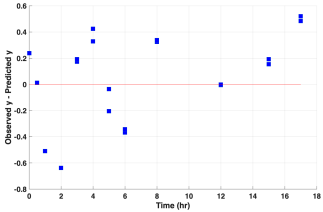

Fig 7. Residual scatter

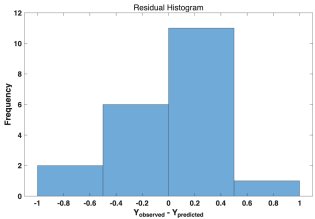

Fig 8. Residual histogram

5 Assumptions:

- ✓ Zero mean (≈ 0.064)
- △ Constant variance
- △ Uncorrelated (runs $= 6 < 10.5$)
- ✓ Normality
- ✓ Correct model

SSC Convergence —Round 2 Before & After Estimation

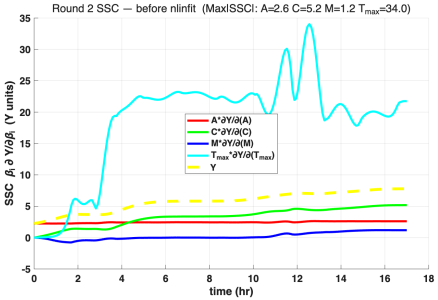

Fig 9a. SSC using Round 1 estimates as initial guess

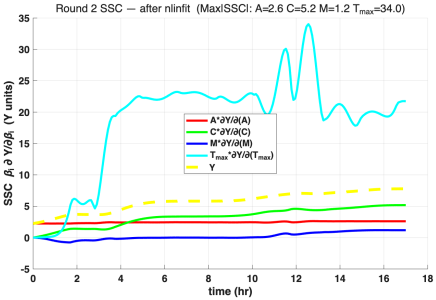

Fig 9b. SSC after Round 2 OLS estimation

SSC Convergence Criterion

SSC shapes and magnitudes are consistent before and after Round 2 estimation ⇒ parameter set is **stable and identifiable**; no further reduction needed.

Optimal Experimental Design

- Equally-spaced `linspace` points (consistent with `inv_soln5.m`); unscaled Jacobian $J = \partial Y/\partial \beta$
- Spikes at $n \approx 27, 34, 48$: aliasing —equally-spaced points align with the $T_{\max}$ SSC peak at $t \approx 12$ hr
- For $n < 22$: smooth increase of Δ; A has smallest C_{ii} (most accurately determinable), C most uncertain

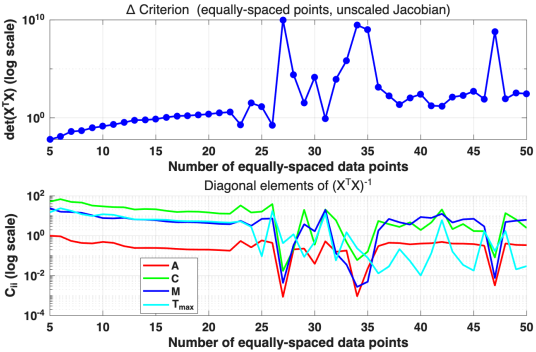

Fig 10. Δ criterion and C_{ii} vs. data points

Bootstrapping of Residuals

1000 bootstrap iterations (residual resampling), 1000/1000 valid.

Param	Bootstrap 95% CI	Asymptotic 95% CI
A	[2.161, 2.511]	[1.985, 2.504]
C	[12.59, 14.61]	[12.29, 14.49]
M	[3.381, 3.612]	[3.479, 3.578]
$T_{\max}$	[41.29, 42.02]	[41.60, 41.70]

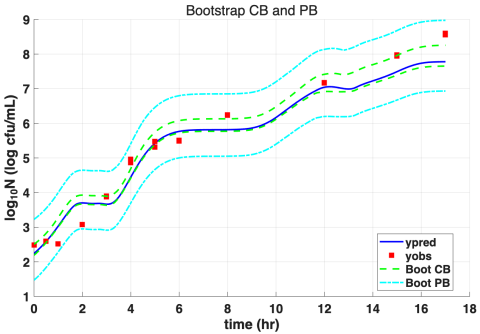

Fig 11. Bootstrap CB and PB

Asymptotic vs Bootstrap Comparison

Metric	Asymptotic	Bootstrap
CB width (avg)	1.044	**0.372**
PB width (avg)	3.032	**1.812**

- Bootstrap CB is $\mathbf{3}\times$ narrower; Bootstrap PB is $\mathbf{1.7}\times$ narrower
- Asymptotic method overestimates uncertainty due to C–$T_{\max}$ ($R = -0.990$) and A–M ($R = -0.956$) correlations inflating the covariance matrix
- Bootstrap provides more realistic uncertainty quantification for this nonlinear problem

Conclusions

1. Modified Gompertz model successfully predicts *Salmonella* growth under sinusoidal temperature
2. **Parameter selection**: SSC analysis + collinearity check $\Rightarrow$ identified $[A, C, M, T_{\max}]$ as estimable via stepwise elimination ($7 \to 5 \to 4$); fixed $T_{\min}$, b (low SSC) and a (collinear with M)
3. Robust OLS fit achieved: cond$(\mathbf{J}) = 293$, relRMSE $= 6.3\%$, $R^2 = 0.972$; residual autocorrelation (runs test: $6 < 10.5$) suggests potential for model improvement
4. Bootstrap CI are narrower than asymptotic CI $\Rightarrow$ asymptotic method overestimates uncertainty due to C–$T_{\max}$ ($R = -0.990$) and A–M ($R = -0.956$) correlations

References I

[1] Lavanya Gumudavelli, Prem Rishi Velugoti, Ramaswamy C Anantheswaran, John D Floros, and Stephen J Knabel.
Dynamic predictive model for growth of *Salmonella Enteritidis* in egg yolk.
Journal of Food Science, 72(7):M254–M262, 2007.